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Implant finder

Abstract

Method and apparatus for locating an implant comprising magnetic material and/or a charge storage device in a living animal. The apparatus may include a magnetic field generator (e.g., an electromagnetic field generator) configured to generate a magnetic field (e.g., an electromagnetic field). The apparatus may include a sensor configured to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field. The magnetic material and/or the charge storage device of the implant may cause changes to the magnetic field as the sensor is moved over the implant. The apparatus may include a computer configured to use the sensor signal to detect a location of the implant.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims the benefit of priority to U.S. Provisional Application Ser. No. 63/327,566, filed on Apr. 5, 2022, which is incorporated herein by reference in its entirety.

BACKGROUND

Field of Invention

(1) The present disclosure relates to apparatuses and method for locating an implant including magnetic material and/or a charge storage device in a living animal.

Discussion of the Background

(2) Implants, such as sensors, may be implanted within a living animal (e.g., a human). An implant may, for example, detect the presence or amount of an analyte (e.g., glucose or oxygen) in a medium (e.g., blood or interstitial fluid) within the living animal. Some implants may be relatively

small (e.g., 2-4 mm diameter capsule) and may be implanted in the subcutaneous tissue of a host. The implant may need to be located (e.g., for removal of the implant). Due to the small size of the implant, it may be difficult to locate and retrieve the implant from the subcutaneous tissue of the host.

SUMMARY

(3) One aspect of the invention may provide an apparatus for locating an implant including magnetic material. The apparatus may include a magnetic field generator, a sensor, and a computer. The magnetic field generator may be configured to generate a magnetic field. The sensor may be configured to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field. The magnetic material of the implant may cause changes to the magnetic field as the sensor is moved over the implant. The computer may be configured to use the sensor signal to detect a location of the implant.

(4) In some aspects, the magnetic field generator may include a cylindrical magnet. In some aspects, the cylindrical magnet may be hollow.

(5) In some aspects, the magnetic field generator may include one or more magnets. In some aspects, the magnetic field generator may include two or more magnets. In some aspects, the magnetic field generator may further include a housing configured to hold the two or more magnets. In some aspects, the magnetic field generator may include four magnets. In some aspects, the magnetic field generator may include six magnets. In some aspects, the one or more magnets may include one or more permanent magnets. In some aspects, the one or more magnets may include one or more electromagnets.

(6) In some aspects, the magnetic field generated by the magnetic field generator may be a substantially uniform magnetic field.

(7) In some aspects, the magnetic field may be substantially symmetric about a longitudinal axis at the center of the magnetic field generator. In some aspects, the magnetic field may be non-uniform and/or asymmetric about a longitudinal axis at the center of the magnetic field generator. In some aspects, the sensor may be located along or offset from the longitudinal axis at the center of the magnetic field generator.

(8) In some aspects, the computer may be configured to, in detecting the location of the implant, determine edges of the magnetic material of the implant based on locations of bimodal peaks in the changes in the magnetic field during movement of the sensor along a longitudinal axis of the implant. In some aspects, the computer may be configured to determine edges of the implant based on the determined edges of the magnetic material of the implant and one of more offsets between edges of the implant and edges of the magnetic material of the implant. In some aspects, the computer may be configured to determine a depth of the implant based on a magnitude of the change in the magnetic field at the bimodal peaks in the changes in the magnetic field during the movement of the sensor along the longitudinal axis of the implant. In some aspects, the computer may be configured to determine an orientation of the implant based on a difference between magnitudes of the change in the magnetic field at the bimodal peaks in the changes in the magnetic field during the movement of the sensor along the longitudinal axis of the implant.

(9) In some aspects, the computer may be configured to calculate a derivative of the changes in the magnetic field and to use the calculated derivative to detect the location of the implant. In some aspects, the derivative of the changes in the magnetic field may be with respect to time (during the movement of the sensor along the longitudinal axis of the implant). In some alternative aspects, the derivative of the changes in the magnetic field may be with respect to the position of the sensor (during movement of the sensor along the longitudinal axis of the implant). In some aspects, the computer may be configured to determine edges of the magnetic material of the implant based on locations where the derivative of the changes in the magnetic field equals zero during movement of the sensor along a longitudinal axis of the implant. In some aspects, the computer may be configured to determine a depth of the implant based on magnitudes of the changes in the magnetic

field at the locations where the derivative of the changes in the magnetic field equals zero during the movement of the sensor along the longitudinal axis of the implant. In some aspects, the computer may be configured to determine an orientation of the implant based on a difference between magnitudes of the changes in the magnetic field at the locations where the derivative of the changes in the magnetic field equals zero during the movement of the sensor along the longitudinal axis of the implant. In some aspects, the computer may be configured to determine a midline of the implant based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor across a longitudinal axis of the implant. In some aspects, the derivative of the changes in the magnetic field may be with respect to time or position (as the sensor is moved across the longitudinal axis of the implant).

(10) In some aspects, the computer may be further configured to use the one or more sensor signals to detect an orientation of the implant.

(11) In some aspects, the apparatus may further include a display, and the computer may be configured to cause the display to display an indication of the detected location of the implant. In some aspects, the display may be located above the sensor. In some aspects, the indication of the detected location of the implant may include an implant image, and a location of the implant image on a screen of the display relative to a point on the screen of the display may correspond to the detected location of the implant relative to the sensor. In some aspects, the implant image may have an orientation that corresponds to a detected orientation of the implant.

(12) In some aspects, the apparatus may further include an incision marking tool configured to identify an incision location for removing the implant.

(13) In some aspects, a diameter of the magnetic field generator may be equal to the length of the implant.

(14) In some aspects, the apparatus may further include a position detector configured to generate a location signal indicative of a location of the sensor on the skin surface. In some aspects, the position detector may include a motion detector configured to detect movement of the sensor and to generate a motion signal indicative of the detected movement of the sensor, and the location signal may include the motion signal. In some aspects, the computer may be configured to use the sensor signal and the location signal to generate a map of sensor signals at different locations of the sensor on the skin surface.

(15) Another aspect of the invention may provide a method for locating an implant including magnetic material. The method may include using a magnetic field generator to generate a magnetic field. The method may include using a sensor to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field. The magnetic material of the implant may cause changes to the magnetic field as the sensor is moved over the implant. The method may include using a computer to detect a location of the implant based on the sensor signal.

(16) In some aspects, detecting the location of the implant may include determining edges of the magnetic material of the implant based on locations of bimodal peaks in the changes in the magnetic field during movement of the sensor along a longitudinal axis of the implant.

(17) Yet another aspect of the invention may provide a method for locating an implant including magnetic material. The method may include moving an apparatus including a magnetic field generator and a sensor across a longitudinal axis of the implant. The magnetic material of the implant may cause changes in a magnetic field generated by the magnetic field generator as the apparatus is moved across the longitudinal axis of the implant, and the sensor may detect changes in the magnetic field. The method may include determining a midline of the implant based on a location where the changes in the magnetic field are greatest as the apparatus is moved across the longitudinal axis of the implant. The method may include moving the apparatus along the determined midline of the implant. The method may include determining edges of the magnetic material of the implant based on locations of bimodal peaks in the changes in the magnetic field as the apparatus is moved along the determined midline of the implant.

(18) In some aspects, the method may further include using an incision marking tool of the apparatus to mark an incision location for removing the implant.

(19) Still another aspect of the invention may provide an apparatus for locating an implant including a charge storage device. The apparatus may include an electromagnetic field generator, a sensor, and a computer. The electromagnetic field generator may be configured to generate an electromagnetic field. The sensor may be configured to detect changes in the electromagnetic field and to generate a sensor signal indicative of the changes in the electromagnetic field. The charge storage device of the implant may cause changes to the electromagnetic field as the sensor is moved over the implant. The computer may be configured to use the sensor signal to detect a location of the implant.

(20) Yet another aspect of the invention may provide a method for locating an implant including a charge storage device. The method may include using an electromagnetic field generator to generate an electromagnetic field. The method may include using a sensor to detect changes in the electromagnetic field and to generate a sensor signal indicative of the changes in the electromagnetic field. The charge storage device of the implant may cause changes to the electromagnetic field as the sensor is moved over the implant. The method may include using a computer to detect a location of the implant based on the sensor signal.

(21) Still another aspect of the invention may provide a method for locating an implant including a charge storage device. The method may include moving an apparatus including an electromagnetic field generator and a sensor across a longitudinal axis of the implant. The charge storage device of the implant may cause changes in an electromagnetic field generated by the electromagnetic field generator as the apparatus is moved across the longitudinal axis of the implant, and the sensor may detect changes in the electromagnetic field. The method may include determining a midline of the implant based on a location where the changes in the electromagnetic field are greatest as the apparatus is moved across the longitudinal axis of the implant. The method may include moving the apparatus along the determined midline of the implant. The method may include determining edges of the charge storage device of the implant based on locations of bimodal peaks in the changes in the electromagnetic field as the apparatus is moved along the determined midline of the implant.

(22) These and other embodiments encompassed within the systems and methods are described in the detailed description of the invention below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and form part of the specification, illustrate various, non-limiting embodiments of the present invention. In the drawings, like reference numbers indicate identical or functionally similar elements.

(2) FIG. 1 is a block diagram illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

(3) FIG. 2A is a cross-sectional side view illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

(4) FIG. 2B is a cross-sectional top view illustrating a non-limiting example of an implant finder embodying aspects of the present invention.

(5) FIGS. 2C and 2D are two-dimensional contour plot slices in the XY and ZY planes, respectively, illustrating the magnitude of a magnetic field generated by a non-limiting example of a hollow, cylindrical magnetic field generator embodying aspects of the present invention.

(6) FIG. 2E is a cross-sectional top view illustrating a non-limiting example of an implant finder embodying aspects of the present invention.

(7) FIGS. 2F and 2G are two-dimensional contour plot slices in the XY and ZY planes, respectively, illustrating the magnitude and vector of a magnetic field generated by a non-limiting example of a magnetic field generator including six magnets embodying aspects of the present invention.

(8) FIGS. 2H and 2I are two-dimensional contour plot slices in the XY and ZY planes, respectively, illustrating the magnitude and vector of a magnetic field generated by a non-limiting example of a magnetic field generator including four magnets embodying aspects of the present invention.

(9) FIG. 3A is a cross-sectional side-view illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

(10) FIGS. 3B and 3C are two-dimensional contour plot slices in the XY and ZY planes, respectively, illustrating the magnitude and vector of a magnetic field generated by a non-limiting example of a cylindrical magnetic field generator embodying aspects of the present invention.

(11) FIGS. 4A and 4B illustrate a non-limiting example of a magnetometer including the sensor of an implant finder embodying aspects of the present invention.

(12) FIGS. 5A and 5B are perspective views illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

(13) FIG. 5C is a see-through perspective view illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

(14) FIGS. 5D and 5E are perspective and top views illustrating a non-limiting example of an implant and magnetic field generator and sensor of an implant finder embodying aspects of the present invention.

(15) FIG. 5F is a see-through perspective view illustrating a non-limiting example of an implant and magnetic field generator and sensor of an implant finder embodying aspects of the present invention.

(16) FIG. 6 is a block diagram illustrating a non-limiting example of a computer of an implant finder embodying aspects of the present invention.

(17) FIG. 7A is a cross-sectional side view illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

(18) FIGS. 7B-7F illustrate non-limiting examples of a display screen of the user interface of an implant finder embodying aspects of the present invention.

(19) FIGS. 8A and 8B are perspective views illustrating a non-limiting example of an implant embodying aspects of the present invention.

(20) FIGS. 9A and 9B illustrate movement of the implant finder across a longitudinal axis of an implant that is not tilted relative to the skin surface and a magnitude of changes to a magnetic field during the movement, respectively, embodying aspects of the present invention.

(21) FIGS. 9C and 9D illustrate movement of the implant finder along a longitudinal axis of the implant that is not tilted relative to the skin surface and a magnitude of changes to a magnetic field during the movement, respectively, embodying aspects of the present invention.

(22) FIG. 9E illustrates an identified implant edge and marked incision location embodying aspects of the present invention.

(23) FIGS. 10A and 10B illustrate movement of the implant finder across a longitudinal axis of a downward-tilted implant and a magnitude of changes to a magnetic field during the movement, respectively, embodying aspects of the present invention.

(24) FIGS. 10C and 10D illustrate movement of the implant finder along a longitudinal axis of the downward-tilted implant and a magnitude of changes to a magnetic field during the movement, respectively, embodying aspects of the present invention.

(25) FIG. 10E illustrates an identified implant edge and marked incision location embodying aspects of the present invention.

(26) FIGS. 11A and 11B illustrate movement of the implant finder across a longitudinal axis of an implant and a magnitude of changes to a magnetic field during the movement for different

distances between the sensor and implant, respectively, embodying aspects of the present invention.

(27) FIGS. **12A** and **12B** illustrate movement of the implant finder along a longitudinal axis of an implant and a magnitude of changes to a magnetic field during the movement for different distances between the sensor and implant, respectively, embodying aspects of the present invention.

(28) FIGS. **13A** and **13B** illustrate movement of the implant finder in a diagonal direction relative to a longitudinal axis of an implant and a magnitude of changes to a magnetic field during the movement, respectively, embodying aspects of the present invention.

(29) FIGS. **13C-13E** illustrate X-, Y-, and Z-components, respectively, of the magnetic field change as the implant finder is moved in the diagonal direction relative to the longitudinal axis of the implant embodying aspects of the present invention.

(30) FIGS. **14A-14C** illustrate an implant twisted out-of-plane, changes to a magnetic field during movement of the implant finder across a longitudinal axis of the implant, and changes to a magnetic field during movement of the implant finder along a longitudinal axis of the implant, respectively, embodying aspects of the present invention.

(31) FIGS. **15A-15C** illustrate movement of the implant finder to identify an orientation of an implant, movement of the implant finder along a longitudinal axis of the implant, and an identified implant edge and marked incision location, respectively, embodying aspects of the present invention.

(32) FIGS. **16A-16C** illustrate identification of an approximate location of an implant, markings indicating a course for movement of an implant finder, and an identification of implant edge and incision coordinates, respectively, embodying aspects of the present invention.

(33) FIG. **17** is a flowchart illustrating a non-limiting example of a process for locating an implant including magnetic material embodying aspects of the present invention.

(34) FIG. **18** is a flowchart illustrating a non-limiting example of a process for locating an implant including magnetic material embodying aspects of the present invention.

(35) FIG. **19** is a side view illustrating a non-limiting example of an implant embodying aspects of the present invention.

(36) FIG. **20** is a block diagram illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

(37) FIG. **21A** is a cross-sectional side view illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

(38) FIG. **21B** is a cross-sectional top view illustrating a non-limiting example of an implant finder embodying aspects of the present invention.

(39) FIG. **21C** is a perspective view illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

(40) FIG. **22** is a flowchart illustrating a non-limiting example of a process for locating an implant including a charge storage device embodying aspects of the present invention.

(41) FIG. **23** is a flowchart illustrating a non-limiting example of a process for locating an implant including a charge storage device embodying aspects of the present invention.

DETAILED DESCRIPTION

(42) FIG. **1** is a block diagram illustrating an implant **100** and an implant finder **101** embodying aspects of the present invention. In some aspects, the implant finder **101** may be for locating the implant **100**, which may have been implanted in tissue below the surface of skin of a living animal. In some aspects, as shown in FIG. **1**, the implant **100** may include magnetic material **124** (e.g., a magnetic core). In some aspects, the magnetic material **124** may include, for example and without limitation, a ferromagnetic material (e.g., iron) or a ferrimagnetic material (e.g., ferrite).

(43) In some aspects, the implant finder **101** may include a magnetic field generator **103**, a sensor **105**, a computer **106**, a battery **117**, a wireless communication integrated circuit (IC) **119**, a connector **121**, a connector IC **123**, a charger IC **125**, a user interface **127**, and/or a position detector **137**. In some aspects, the user interface **127** of the implant finder **101** may include a

display **129** (e.g., an optical display such as, for example, a light emitting diode (LED) display), a speaker **131**, a vibration motor **133**, and/or a user input **135**.

(44) In some aspects, the magnetic field generator **103** may be configured to generate a magnetic field. In some aspects, the sensor **105** may be configured to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field. In some aspects, the magnetic material **124** of the implant **100** may cause changes to the magnetic field as the sensor **105** is moved over the implant **100**. In some aspects, the computer **106** may be configured to use the sensor signal to detect a location of the implant **100**.

(45) In some aspects, as shown in FIGS. 2A, 2B, 2E, and 3A, the magnetic field generator **103** may include one or more magnets **109**. In some aspects, the one or more magnets **109** may be, for example and without limitation, permanent magnets and/or electromagnets (e.g., a coil of wire wound around a magnetic core made from a ferromagnetic or ferrimagnetic material). In some aspects, as shown in the FIGS. 5A-5C, the magnetic field generator **103** may include a housing **107**, and the housing **107** may be configured to hold the one or more magnets **109**. However, a housing **107** is not required, and, in some alternative aspects (e.g., some aspects in which the magnetic field generator **103** consists of only a single magnet **109**), the magnetic field generator **103** does not include a housing **107**.

(46) In some aspects, as shown in FIGS. 2A, 2B, and 2E, the sensor **105** may be located on a longitudinal axis at the center of the magnetic field generator **103**. However, this is not required, and, in some alternative aspects, the sensor **105** is not located on a longitudinal axis at the center of the magnetic field generator **103**. For example, as shown in FIG. 3A, the sensor **105** may be located adjacent to the magnet **109** of the magnetic field generator **103** (as opposed to at the center of one or more magnets **109** of the magnetic field generator **103**).

(47) In some aspects, as shown in FIG. 2B, the magnetic field generator **103** may include a hollow cylindrical magnet **109**. In some alternative aspects, as shown in FIG. 2E, the magnetic field generator **103** may include two or more magnetics **109** (e.g., four or six magnets). In some other alternative aspects, as shown in FIG. 3A, the magnetic field generator **103** may include a solid cylindrical magnet **109**.

(48) In some aspects, the one or more magnets **109** of the magnetic field generator **103** may generate a substantially uniform magnetic field. In some aspects, the hollow cylindrical magnet **109** of the magnetic field generator **103** illustrated in FIG. 2B may generate a magnetic field having the magnitudes shown in the two-dimensional contour plot slices of FIGS. 2C and 2D. In some aspects, the six magnets **109** of the magnetic field generator **103** illustrated in FIG. 2E may generate a magnetic field having the magnitudes shown in the two-dimensional contour plot slices of FIGS. 2F and 2G. In some aspects, the magnetic field generator **103** may include four magnets **109**, and the four magnets **109** may generate a magnetic field having the magnitudes shown in the two-dimensional contour plot slices of FIGS. 2H and 2I. In some aspects, the solid cylindrical magnet **109** of the magnetic field generator **103** illustrated in FIG. 3A may generate a magnetic field having the magnitudes shown in the two-dimensional contour plot slices of FIGS. 3B and 3C. In some aspects, as shown in FIGS. 2C, 2D, 2F-2I, 3B, and 3C, the magnetic field generated by the one or more magnetics **109** of the magnetic field generator **103** may be substantially symmetric about a longitudinal axis at the center of the magnetic field generator **103**. However, this is not required, and in some alternative aspects, the magnetic field may have a different profile (e.g., the magnetic field may be non-uniform and/or asymmetric about the longitudinal axis at the center of the magnetic field generator). In some aspects, as shown in FIGS. 2A, 2B, and 2E, the sensor **105** may be located on the longitudinal axis at the center of the magnetic field generator **103**.

(49) In some aspects, the sensor **105** may be a magnetic field sensor. In some aspects, the sensor **105** may be a tunnel magnetoresistance (TMR) magnetic field sensor (e.g., a TMR full bridge magnetic field sensor) or a Hall sensor (e.g., a high field Hall sensor). In some aspects, as shown in FIGS. 4A and 4B, the sensor **105** may be part of a magnetometer **400** (e.g., an axial low-field

magnetometer). In some aspects, the magnetometer **400** may include the sensor **105**, a probe **402** (e.g., a rigid probe), a cable **404** (e.g., a flexible cable), and data acquisition electronics **406**, and/or a connector **408** (e.g., a USB connector). In some aspects, the connector **408** of the magnetometer **400** may be connected to the computer **106** of the implant finder **101**.

(50) In some aspects, as shown in FIGS. 5A-5F, the implant finder **101** may include the magnetic field generator **103** and the sensor **105**. In some aspects, the sensor **105** may be located on the longitudinal axis at the center of the magnetic field generator **103**. In some aspects, the magnetic field generator **103** may include a housing **107**. In some aspects, the housing **107** may include housing bodies **115** (e.g., polymer bodies). In some aspects, as shown in FIGS. 5C-5F, the magnetic field generator **103** may include one or more magnets **109**. In some aspects, the housing **107** may be configured to house the one or more magnets **109**. In some aspects, as shown in FIGS. 5C-5F, the magnetic field generator **103** may include six magnets **109**. However, in some alternative aspects, the magnetic field generator **103** may include a different number of magnets **109** (e.g., ten, nine, eight, seven, five, four, three, or two magnets or a single magnet, such as a hollow cylindrical magnet). In some aspects, as shown in FIG. 5E, a diameter or width of the magnetic field generator **103** may be equal to the length of the implant **100**. However, this is not required, and, in some alternative aspects, the magnetic field generator **103** may have a different size (e.g., a diameter or width that is less than or greater than the length of the implant **100**).

(51) In some aspects, as shown in FIGS. 5A-5C, the implant finder **101** may include a handle **111**. In some aspects, as shown in FIGS. 5A-5C, the implant finder **101** may include an incision marking tool **113**. In some aspects, the incision marking tool **113** may be configured such that a user may use the incision marking tool **113** to mark an appropriate location on the skin surface **154** for an incision that would enable removal of the implant **100**. In some aspects, the user may use the incision marking tool **113** to mark the incision location when the implant finder **101** is located over the implant **100**. In some aspects, the incision marking tool **113** may be configured to mark the skin surface **154** above a determined edge of the implant **100**, a determined center of the implant **100**, a determined edge of the magnetic material **124**, or a determined center of the magnetic material **124**. In some alternative aspects, the incision marking tool **113** may be configured to mark the skin surface **154** above a location offset from a determined edge of the implant **100**, a determined center of the implant **100**, a determined edge of the magnetic material **124**, and/or a determined center of the magnetic material **124**. In some alternative aspects, the incision marking tool **113** may be configured mark the skin surface at a longitudinal axis at the center of the magnetic field generator **103**.

(52) In some aspects in which the implant finder **101** includes the connector **121**, the connector **121** may be, for example and without limitation, a Micro-Universal Serial Bus (USB) connector. In some aspects, the connector **121** may be configured to enable a wired connection to an external device, such as a personal computer or display device. In some aspects, the implant finder **101** may exchange data to and from the external device through the connector **121** and/or may receive power through the connector **121**. In some embodiments, the connector IC **123** may be, for example and without limitation, a USB-IC, which may control transmission and receipt of data through the connector **121**.

(53) In some aspects in which the implant finder **101** includes the battery **117**, the battery **117** may provide operating power for the implant finder **101**. In some aspects, the battery **117** may be a rechargeable battery. In some aspects, the battery **117** may be, for example and without limitation, a lithium-polymer battery. In some aspects, the battery **117** may have a short recharge duration and/or may have a small size. In some aspects in which the display device **105** includes the charger IC **125**, the charger IC **125** may receive power via the connector **121** and charge the battery **117**.

(54) In some aspects in which the implant finder **101** includes the wireless communication IC **119**, the wireless communication IC **119** may enable wireless communication with one or more external devices, such as, for example, one or more personal computers and/or one or more other display

devices (e.g., a smartphone or tablet executing an application). In some aspects, the wireless communication IC **119** may employ one or more wireless communication standards to wirelessly transmit data. The wireless communication standard employed may be any suitable wireless communication standard, such as an IEEE 802.11 standard, an ANT standard, a Bluetooth standard, or a Bluetooth Low Energy (BLE) standard (e.g., BLE **4.0**). In some aspects, the wireless communication IC **119** may include an antenna (e.g., a Bluetooth antenna, a Wi-Fi antenna, and/or one or more cellular antennas). In some aspects, the antenna of the wireless communication IC **119** may be entirely contained within a housing (e.g., housing **107**) of the implant finder **101**. However, this is not required, and, in alternative embodiments, all or a portion of the antenna of the wireless communication IC **119** may be external to a housing of the implant finder **101**.

(55) In some aspects in which the implant finder **101** includes the position detector **137**, the position detector **137** may be configured to generate a location signal indicative of a location of the sensor on the skin surface **154**. In some aspects, the position detector **137** may, for example and without limitation, include a motion detector configured to detect movement of the sensor **105** (and/or implant finder **101**) relative to the skin surface **154** and to generate a motion signal indicative of the detected movement of the sensor **105**, and the location signal may include the motion signal. In some aspects, the motion detector may include a mechanical motion detector (e.g., that makes use of a motion of a ball inside the mechanical motion detector) and/or an optical motion detector (e.g., that uses light to detect movement of the sensor **105**).

(56) In some aspects in which the implant finder **101** includes the user interface **127**, the user interface **127** may include one or more of a display **129** and a user input **135**. In some aspects, the display **129** may include a liquid crystal display (LCD) and/or a light emitting diode (LED) display. In some aspects, the user input **135** may include one or more buttons, a keyboard, a keypad, and/or a touchscreen. In some aspects, the user interface **127** may include one or more of a speaker **131** (e.g., a beeper) and a vibration motor **133**, which may be activated, for example, in the event that a condition is met (e.g., detection of an edge of the implant **100** and/or a detection of a midline of the implant **100**).

(57) In some aspects in which the implant finder **101** includes the computer **106**, the computer **106** may control the overall operation of the implant finder **101**. For example, the implant finder **101** may control the wireless communication IC **119**, the connector IC **123**, the charger IC **125**, the position detector **137**, the magnetic field generator **103**, the sensor **105**, and/or the user interface **127**. In some aspects, the computer **106** may receive and/or process data from the sensor **105**, the position detector **137**, and/or user input **135** of the user interface **127**. For example, in some aspects, the computer **106** may be configured to use the sensor signal generated by the sensor **105** to detect a location of the implant **100**. In some aspects, the computer **106** may be configured to use the sensor signal generated by the sensor **105** and the location signal generated by the position detector **137** to generate a map of sensor signals at different locations of the sensor **105** on the skin surface **154**. In some aspects, the computer **106** may additionally or alternatively control the display **129**, speaker **131**, and/or vibration motor **135** to provide information about the detected location of the implant **100**. For example, in some aspects, the computer **106** may cause the display **129** to display a notification (e.g., one or more optical displays produced by one or more LEDs), the speaker **131** to beep, and/or the vibration motor **135** to vibrate when the computer **106** detects an edge of the implant **100** and/or a midline of the implant **100**. In some aspects, the computer **106** may additionally or alternatively be configured to cause the display **129** to display an indication of the detected location of the implant **100**.

(58) FIG. **6** is a block diagram of the computer **106** of the implant finder **101** according to some aspects. In some aspects, the computer **106** can be adapted to perform any of methods, processes, or steps disclosed herein. As shown in FIG. **6**, the computer **106** may include processing circuitry (PC) **302**, which may include one or more processors (P) **355** (e.g., one or more general purpose microprocessors and/or one or more other processors, such as an application specific integrated

circuit (ASIC), field-programmable gate arrays (FPGAs), and the like).

(59) In some aspects, as shown in FIG. 6, the computer **106** may include one or more storage units (a.k.a., “data storage systems”) **308** which may be co-located or geographically distributed and which may include one or more non-volatile storage devices and/or one or more volatile storage devices. In some aspects where the PC **302** includes a programmable processor, the one or more storage units may include a computer program product (CPP) **341**. In some aspects, the CPP **341** may include a computer readable medium (CRM) **342** storing a computer program (CP) **343** comprising computer readable instructions (CRI) **344**. In some aspects, the CRM **342** may be a non-transitory computer readable medium, such as, magnetic media (e.g., a hard disk), optical media, memory devices (e.g., random access memory, flash memory), and the like. In some aspects, the CRI **344** of the computer program **343** is adapted such that when executed by the PC **302**, the CRI **341** may cause the computer **106** to perform steps described herein (e.g., steps described herein with reference to the flow charts). In other aspects, the computer **106** may be adapted to perform steps described herein without the need for code. That is, for example, the PC **302** may consist merely of one or more ASICs. Hence, the features of the aspects described herein may be implemented in hardware and/or software.

(60) In some aspects, as shown in FIG. 7A, the display **129** of the user interface **127** of the implant finder **101** may be located above the sensor **105**. In some aspects, the magnetic field generator **103** may be configured to hold the display **129** in place at the top of the implant finder **101**. In some aspects, as shown in FIG. 7A, the housing **107** of the magnetic field generator **103** may be configured to house the one or more magnets **109** of the magnetic field generator **103**, the computer **106**, and/or the display **129**. However, this is not required, and, in some alternative aspects, the display **129** may be located above the magnetic field generator **103**.

(61) In some aspects, as shown in FIGS. 7B-7F, the display **129** of the user interface **127** may include a display screen **702**. In some aspects (e.g., in some aspects in which the display **129** is located above the sensor **105** and the computer **106** is configured to cause the display **129** to display an indication of the detected location of the implant **100**), the indication of the detected location of the implant **100** may include an implant image **704**, and a location of the implant image **704** on the screen **702** of the display **129** relative to a point on (e.g., the center of) the screen **702** of the display **129** may correspond to the detected location of the implant **100** relative to the sensor **105**. In some aspects, as shown in FIGS. 7B-7E, the implant image **704** may have an orientation that corresponds to an orientation of the implant **100** detected by the computer **106** using sensor signals generated by the sensor **105**.

(62) In some aspects, as shown in FIG. 7C, depending on the location of the implant **100** relative to the sensor **105**, the implant image **704** may be a partial implant image. In some aspects, as shown in FIGS. 7B-7F, a mark **706** may identify the point on the screen **702** that corresponds to the location of the sensor **105**. In some aspects (e.g., some aspects in which the screen **702** of the display **129** is larger than and/or extends beyond the magnetic field generator **103**), as shown in FIG. 7E, the computer **106** may cause the display **129** to display a magnetic field generator image **708** indicative of the location of the outer perimeter of the magnetic field generator **103**. In some aspects, the location of the implant image **704** relative to the magnetic field generator image **708** may correspond to the detected location of the implant **100** relative to the location of magnetic field generator **103**. In some aspects, as shown in FIG. 7F, the computer **106** may cause the display **129** to display a target region **710**, and the implant finder **101** may be properly positioned over the implant **100** such that the incision location can be marked (e.g., using the incision marking tool **113**) when the implant image **704** is located within the target region **710**.

(63) FIGS. 8A and 8B illustrate a non-limiting example of an implant **100** that may be located by the implant finder **101**. In some aspects, the implant **100** may be a small, fully subcutaneously implantable sensor, e.g., that measures the amount or concentration of an analyte (e.g., glucose) in a medium (e.g., interstitial fluid) of a living animal (e.g., a living human). In some alternative

aspects, the implant **100** is not a sensor and may instead be a different type of implantable device, such as, for example and without limitation, a tag, an insulin pump, a pacemaker, or an electrical/heat therapy device.

(64) In some aspects, as illustrated in FIG. **8A**, the implant **100** may be implanted in the tissue **150** (e.g., subcutaneous tissue) of the living animal, where the implant **100** may rest in a pocket **152** in the tissue **150** below the skin surface **154**, and the pocket **152** may surround the implant **100**. In some aspects, the pocket **152** may be created by a tissue dissector tool before implantation of the implant **100** or by the implantation process.

(65) In some aspects, as illustrated in FIG. **8A**, the implant **100** may include a housing **102** and one or more analyte indicators **104** coated, diffused, adhered, embedded, or grown on or in at least a portion of the exterior surface of the housing **102**. In some aspects, the one or more analyte indicators **104** (e.g., polymer grafts or hydrogels) of the implant **100** may include one or more indicator molecules (e.g., fluorescent indicator molecules). In some aspects, the indicator molecules may produce (e.g., exhibit) one or more detectable properties (e.g., optical properties) that vary in accordance with the amount or concentration of the analyte in proximity to an analyte indicator **104**. In some aspects, the indicator molecules may emit an amount of emission light (e.g., fluorescent light) that varies in accordance with the amount or concentration of the analyte in proximity to the analyte indicator **104**.

(66) In some embodiments, the implant **100** may include a substrate **110** (e.g., a printed circuit board (PCB) or flexible PCB), one or more light sources **111** mounted on or fabricated in the substrate **110**, and one or more photodetectors **112**, **114**, and **116** (e.g., photodiodes, phototransistors, photoresistors, or other photosensitive elements) mounted on or fabricated in the substrate **110**. In some embodiments, the light source **111** may be configured to emit an excitation light over an excitation wavelength range that interacts with the one or more indicator molecules in the analyte indicator **104**. In some aspects, one or more of the photodetectors **112**, **114**, and **116** may output a signal indicative of an amount of light received by the photodetector. In some examples, the signal output by the one or more internal photodetectors **112**, **114**, and **116** may be indicative of an amount or concentration of an analyte in a medium in proximity to the analyte indicator **104**.

(67) In some aspects, as shown in FIGS. **8A** and **8B**, the implant **100** may include an inductor **120**, which may be, for example, a ferrite based micro-antenna. In some aspects, the inductor **120** may include a conductor **122** in the form of a coil and a magnetic material **124** in form of a magnetic core **124**. In some aspects, the core **124** may be, for example and without limitation, a ferrite core. In some aspects, the inductor **120** may be connected to circuitry (e.g., an application specification integrated circuit (ASIC)) of the implantable device **100**. In some aspects, the inductor **120** may communicate with an external device (not shown) by, passive telemetry (e.g., near field communication), such that power and/or data is transferred between the implant **100** and the external device.

(68) In some aspects, the implant finder **101** may be used for locating an implant **100** (e.g., an implant **100** that has been implanted in tissue **150** below a skin surface **154** of a living animal). In some aspects, a first process for locating the implant **100** may include a first step of determining a midline of the implant **100**. In some aspects, determining the midline of the implant **100** may include, as shown in FIGS. **9A**, **10A**, and **11A**, moving the implant finder **101** across (e.g., transverse to) a longitudinal axis of the implant **100**. FIGS. **9A** and **10A** illustrate the movement of the implant finder **101** across the longitudinal axis of an implant **100** that is not tilted relative to the skin surface and across the longitudinal axis of a downward-tilted implant **100**, respectively. In some aspects, the orientation of the implant **100** may be assumed (e.g., the longitudinal axis of an implant **100** implanted in an arm of a living animal may be assumed to be approximately along the long axis of the arm), and the movement of the implant finder **101** across the longitudinal axis of the implant **100** may be transverse to the assumed orientation of the implant **100**. In some aspects,

the orientation of the implant **100** may be determined using records regarding the implantation of the implant **100**, and the movement of the implant finder **101** across the longitudinal axis of the implant **100** may be transverse to the orientation of the implant **100** determined using the implantation records.

(69) In some aspects, locating the implant **100** may include using implantation information for the implant **100**. In some aspects, the implantation information may include implantation location information identifying the location at which the implant **100** was implanted, and the implantation location information may be used to identify and/or limit an initial search area for the implant **100**. In some aspects, the implantation information may additionally or alternatively include implantation depth information identifying the depth at which the implant was implanted, and the implantation depth information may be used to estimate an expected signal range. In some aspects in which there is an offset between the center of the magnetic material **124** and the center of the implant **100**, the implantation information may additionally or alternatively include offset direction information identifying a direction of the offset of the center of the magnetic material **124** relative to the center of the implant **100**, which may be used for identifying edges of the implant **100** and/or an incision location.

(70) In some aspects, the magnetic material **124** of the implant **100** may cause changes in a magnetic field generated by the magnetic field generator **103** of the implant finder **101** as the implant finder **101** is moved across the longitudinal axis of the implant **100**, and the sensor **105** of the implant finder **101** may detect changes in the magnetic field. FIG. **9B** illustrates changes in the magnetic field caused by the magnetic material **124** of the implant **100** that is not tilted relative to the skin surface as detected by the sensor **105** during movement across the longitudinal axis of the implant **100** according to some aspects. FIGS. **10B** and **14B** illustrate changes in the magnetic field caused by the magnetic material **124** of an out-of-plane or downward-tilted implant **100** as detected by the sensor **105** during movement across the longitudinal axis of the implant **100** according to some aspects. In some aspects, as shown in FIGS. **9B**, **10B**, and **14B**, the change in the magnetic field as the implant finder **101** is moved across the longitudinal axis of the implant **100** may be greatest at the location of the midline of the implant **100**.

(71) In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may be configured to determine the midline of the implant **100** based on a location where the changes in the magnetic field are greatest as the implant finder **101** is moved across the longitudinal axis of the implant **100**. In some aspects, the computer **106** of the implant finder **101** may be configured to determine a derivative of the detected changes in the magnetic field, and the computer **106** may be configured to determine the midline of the implant **100** (and the location where the changes in the magnetic field are greatest) based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor **105** across a longitudinal axis of the implant **100**. In some aspects, the derivative of the changes in the magnetic field may be with respect to time (as the sensor **105** is moved across the longitudinal axis of the implant **100**). In some alternative aspects, the derivative of the changes in the magnetic field may be with respect to the position of the sensor (as the sensor **105** is moved across the longitudinal axis of the implant **100**).

(72) In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may cause the user interface **127** to indicate when the implant finder **101** (e.g., the computer **106**) has detected the midline of the implant **100** (e.g., when the computer **106** determines the derivative of the changes in the magnetic field equals zero during movement of the sensor **105** across the longitudinal axis of the implant **100**). In some aspects, the user interface **127** may indicate detection of the midline using one or more of the display **129** (e.g., by displaying a visual indicator of midline detection), speaker **131** (e.g., by emitting an audible sound such as, for example and without limitation, a beep), and vibration motor **133** (e.g., by vibrating).

(73) FIG. **11B** illustrates detected magnetic field changes as the implant finder **101** is moved across the longitudinal axis of the implant **100** for implants **100** at different heights of the sensor **105** of

the implant finder **101** relative to the implant **100**. As shown in FIG. **11B**, the magnitudes of the detected magnetic field changes increase as the height decreases. Accordingly, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may be configured to use the magnitude of the detected magnetic field change at the midline (e.g., the greatest detected magnetic field change during movement of the sensor **105** across the longitudinal axis of the implant **100**) to determine the depth of the implant **100** in the tissue **150**.

(74) In some aspects, the first process for locating the implant **100** may include a second step of determining one or more edges of the magnetic material **124** of the implant **100** and/or a center of the magnetic material **124** of the implant **101**. In some aspects, determining the edge(s) and/or center of magnetic material **124** of the implant **100** may include, as shown in FIGS. **9C**, **10C**, **12A**, and **14A**, moving the implant finder **101** along a longitudinal axis of the implant **100**. FIG. **9C** illustrated the movement of the implant finder **101** along the longitudinal axis of an implant **100** that is not tilted relative to the skin surface. FIGS. **10C** and **14A** illustrate the movement of the implant finder **101** along the longitudinal axis of a downward-tilted or out-of-plane implant **100**. In some aspects, the determined midline may be used to move the implant finder **101** along the longitudinal axis of the implant **100**. For example, in some aspects, moving the implant finder **101** along the longitudinal axis of the implant **100** may include moving implant finder **101** on the skin surface **154** along the determined midline of the implant **100**. In some aspects, the movement along the longitudinal axis of the implant **100** may be parallel to the assumed orientation of the implant **100**. In some aspects, the movement along the longitudinal axis of the implant **100** may be parallel to the orientation of the implant **100** determined using implantation records.

(75) In some aspects, the magnetic material **124** of the implant **100** may cause changes in a magnetic field generated by the magnetic field generator **103** of the implant finder **101** as the implant finder **101** is moved along the longitudinal axis of the implant **100**, and the sensor **105** of the implant finder **101** may detect changes in the magnetic field. FIG. **9D** illustrates changes in the magnetic field caused by the magnetic material **124** of an implant **100** that is not tilted relative to the skin surface as detected by the sensor **105** during movement along the longitudinal axis of the implant **100** according to some aspects. FIGS. **10D** and **14C** illustrate changes in the magnetic field caused by the magnetic material **124** of an out-of-plane or downward-tilted implant **100** as detected by the sensor **105** during movement along the longitudinal axis of the implant **100** according to some aspects. In some aspects, as shown in FIGS. **9D**, **10D**, and **14C**, the detected changes in the magnetic field as the implant finder **101** is moved along the longitudinal axis of the implant **100** may include bimodal peaks (e.g., local maxima) at the locations of the edges of the magnetic material **124** of the implant **100**. In some aspects, as shown in FIGS. **9D**, **10D**, and **14C**, the changes in the magnetic field may include a local minimum between the bimodal peaks at a location of the center of the magnetic material **124** of the implant **100**.

(76) In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may be configured to determine edges of the magnetic material **124** of the implant **100** based on locations of the bimodal peaks in the changes in the magnetic field. In some aspects, the computer **106** of the implant finder **101** may be configured to determine a derivative of the detected changes in the magnetic field, and the computer **106** may be configured to determine the edges of the magnetic material **124** of the implant **100** (and the locations of the bimodal peaks in the magnetic field changes) based on a locations where the derivative of the changes in the magnetic field equals zero during movement of the sensor **105** along a longitudinal axis of the implant **100**. In some aspects, the derivative of the changes in the magnetic field may be with respect to time (as the sensor **105** is moved along the longitudinal axis of the implant **100**). In some alternative aspects, the derivative of the changes in the magnetic field may be with respect to the position of the sensor (as the sensor **105** is moved along the longitudinal axis of the implant **100**). In some aspects, the edges of the magnetic material **124** of the implant **100** may be offset from the edges of the implant **100**. For example, the edges of the magnetic material **124** of the implant **100** are offset from the

edges of the implant **100**. In some aspects in which the edges of the magnetic material **124** are offset from the edges of the implant **100**, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may be configured to determine one or more edges of the implant **100** using the one or more determined edges of the magnetic material **124** and the offset(s).

(77) In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may be configured to determine a center of the magnetic material **124** of the implant **100** based on the location of a local minimum in the magnetic field changes between the bimodal peaks in the magnetic field changes. In some aspects, the computer **106** of the implant finder **101** may be configured to determine a derivative of the detected changes in the magnetic field, and the computer **106** may be configured to determine the center of the magnetic material **124** of the implant **100** (and the locations of local minimum between the bimodal peaks in the magnetic field changes) based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor **105** along a longitudinal axis of the implant **100**. In some aspects, the center of the magnetic material **124** of the implant **100** may be offset from the center of the implant **100** (i.e., the center of the magnetic material **124** may be different than the center of the implant **100**). For example, in the implant **100** illustrated in FIG. 8A, the center of the magnetic material **124** of the implant **100** is offset from the center of the implant **100** (e.g., by 2.5 mm). In some aspects in which the magnetic material **124** is offset from the center of the implant **100**, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may be configured to determine a center of the implant **100** using the determined center of the magnetic material **124** and the offset. In some aspects, the direction of the offset of the center of the magnetic material **124** relative to the center of the implant **100** may be assumed (e.g., the center of the magnetic material **124** may be assumed to be higher along the long axis of the arm than the center of the implant **100**). In some aspects, the direction of the offset of the center of the magnetic material **124** relative to the center of the implant **100** may be additionally or alternatively confirmed or determined (e.g., using implantation records and/or detected changes in the magnetic field caused by metal and/or circuitry on and/or in the substrate **110**, which may extend from one side of the magnetic material **124** of the implant **100** as shown in FIGS. 8A and 8B).

(78) In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may cause the user interface **127** to indicate when the implant finder **101** (e.g., the computer **106**) has detected the edges and/or center of the magnetic material **124** of the implant **100** (e.g., when the computer **106** determines the derivative of the changes in the magnetic field equals zero during movement of the sensor **105** along the longitudinal axis of the implant **100**). In some aspects, the user interface **127** may indicate detection of the edges and/or center using one or more of the display **129** (e.g., by displaying a visual indicator of edges and/or center detection), speaker **131** (e.g., by emitting an audible sound such as, for example and without limitation, a beep), and vibration motor **133** (e.g., by vibrating).

(79) FIG. 12B illustrates detected magnetic field changes as the implant finder **101** is moved along the longitudinal axis of the implant **100** for implants **100** at different heights of the sensor **105** of the implant finder **101** relative to the implant **100**. As shown in FIG. 12B, the magnitudes of the detected magnetic field change at the bimodal peaks increase as the height decreases. Accordingly, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may be configured to use the magnitude of the detected magnetic field change at one or more of the bimodal peaks (e.g., the local maxima in the detected magnetic field change during movement of the sensor **105** along the longitudinal axis of the implant **100**) to determine the depth of the implant **100** in the tissue **150**. In some aspects, the implant finder **101** may additionally or alternatively use the magnitude of the magnetic field change at the midline of the implant **101** to determine the depth of the implant **100** in the tissue **150**.

(80) In some aspects, as shown in FIG. 9D, the magnitudes of the magnetic field change at the bimodal peaks may be equal (or approximately equal such as, for example and without limitation,

within 2 or 3 oersted (Oe))) for an implant **100** that is not tilted relative to the skin surface and has a longitudinal axis that runs parallel to the skin surface **154**. In some aspects, as shown in FIGS. **10D** and **14C**, the magnitudes of the magnetic field change at the bimodal peaks may be different for an implant **100** having a longitudinal axis that is not parallel to the skin surface **154** (e.g., for a downward-tilted implant **100**). In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may determine an orientation of the implant **100** using the magnitudes of the magnetic field change at the bimodal peaks. In some aspects, determining the orientation may include calculating a difference between the magnitudes of the magnetic field change at the bimodal peaks and calculating an angle of the implant **100** relative to the skin surface **154** based on the calculated difference. In some aspects, determining the orientation of the implant **100** may additionally or alternatively include using the magnitudes of the magnetic field change at the bimodal peaks to calculate depths of the edges of the magnetic material **124** (and/or of the edges of the implant **100**).

(81) In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may be configured to cause the user interface **127** (e.g., the display **129** of the user interface **127**) to provide information indicative of a location and/or orientation of the implant **100**. In some aspects, the information indicative of the location and/or orientation of the implant **100** may include information indicative of a midline of the implant **100**, one or more edges of the magnetic material **124**, one or more edges of the implant **100**, a center of the magnetic material **124**, a center of the implant **100**, a depth of the implant **100** in the tissue **150**, an orientation of the implant **100**, and/or an incision location for removal of the implant **100**. In some aspects, the information indicative of the orientation of the implant **100** may include an indication of the angle of the implant **100**, depths of the edges of the magnetic material **124**, and/or depths of the edges of the implant **100**. In some aspects, the information indicative of the location and/or orientation of the implant **100** may include the implant image **704** (e.g., as shown in FIGS. **7B-7F**). In some aspects, the information indicative of the location and/or orientation of the implant **100** may include one or more plots of the magnetic field changes during the movement/sweeps of the implant finder **101** (e.g., the plots shown in FIGS. **9B**, **9D**, **10B**, **10D**, **13B-13E**, **14B**, **14C**). In some aspects, the plots of may show one or more peaks in the magnetic field change (e.g., depending on whether the movement is across, along, or diagonal to the longitudinal axis of the implant **100**).

(82) In some aspects, as shown in FIGS. **9E** and **10E**, the first process for locating the implant **100** may include a third step of marking a location for an incision to remove the implant **100**. In some aspects, the incision location may be marked using the incision marking tool **113** of the implant finder **101** (e.g., when the sensor **105** of the implant finder **101** is positioned over the center of the implant **100**, over the center of the magnetic material **101** of the implant **100**, and/or with the edges of the magnetic field generator **103** positioned over the edges of the magnetic material **124** of the implant **100**).

(83) In some aspects, as shown in FIG. **13A**, the implant finder **101** may be moved in a diagonal direction (e.g., at a 45 degree angle or at a smaller angle such as, for example and without limitation, a 10, 15, 20, 25, or 30 degree angle) relative to a longitudinal axis of the implant **100** (e.g., when a user is attempting to locate the implant **100** and/or by accident when the user is attempting to move the implant finder **101** along the longitudinal axis). FIG. **13B** illustrates detected magnetic field changes as the implant finder **101** is moved diagonal to the longitudinal axis of the implant **100**. FIGS. **13C-13E** illustrate X-, Y-, and Z-components of the magnetic field change as the implant finder **101** is moved diagonal to the longitudinal axis of the implant **100**. In some aspects, the location of the greatest magnitude of the magnetic field change may be used to determine the midline of the implant **100**. In some aspects, the movement/scan direction of the implant finder **101** may be changed until the orientation of the implant **100** (e.g., the longitudinal axis of the implant **100**) is found.

(84) In some aspects, the implant finder **101** (e.g., a computer **106** of the implant finder **101**) may

be configured to control the user interface **127** (e.g., the display **129**) to output information indicative of the sensor signal generated by the sensor **105** (and therefore indicative of the detected changes in the magnetic field caused by the magnetic material **124** of the implant **100**). In some aspects, a user (e.g., clinician) may use the information indicative of the sensor signal live to map out an approximate location of the implant **100**.

(85) In some aspects, a second process may be used to locate the implant **100**. In some aspects, the second process may be used if the first process (described above with respect to FIGS. **9A-10E**) does not expose the double/bimodal peak pattern present when the implant finder **101** is moved along the longitudinal axis of the implant **100**. In some aspects, the second process may include a first step of finding a “best” orientation of the longitudinal axis of the implant **100**. In some aspects, as shown in FIG. **15A**, the first step may include moving the implant finder **101** at different angles (e.g., at incremental 15 degree angles) until double/bimodal peak pattern is identified.

(86) In some aspects, as shown in FIG. **15B**, the second process may include a second step identifying a center of the magnetic material **124** of the implant **100**. In some aspects, the second step may include moving the implant finder **101** along the longitudinal axis identified in the first step. In some aspects, as shown in FIGS. **9D**, **10D**, and **14C**, the detected changes in the magnetic field as the implant finder **101** is moved along the longitudinal axis of the implant **100** may include bimodal peaks (e.g., local maxima) at the locations of the edges of the magnetic material **124** of the implant **100**. In some aspects, the center of the magnetic material **124** may be identified by finding a local minimum in the magnetic field change between the bimodal peaks in the magnetic field changes. In some aspects, the computer **106** of the implant finder **101** may be configured to determine a derivative of the detected changes in the magnetic field, and the computer **106** may be configured to determine the center of the magnetic material **124** of the implant **100** (and the locations of local minimum between the bimodal peaks in the magnetic field changes) based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor **105** along a longitudinal axis of the implant **100**. In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may be configured to cause the user interface **127** to provide an indication of the center of the magnetic material **124**. In some aspects, the indication of the center of the magnetic material **124** may include a plot of the magnetic field changes during the sweep and/or a visual (e.g., flashing light), aural (e.g., beep), or vibratory indication when a derivative of the magnitude of the magnetic field change is zero between the bimodal peaks).

(87) In some aspects, the second process may include a third step in which the implant finder **101** (e.g., the computer **106** of the implant finder **101**) calculates the depth and/or orientation (e.g., downward angulation) of the implant **100**. In some aspects, the implant finder **101** may calculate the depth and/or orientation of the implant **100** using the magnitudes of the magnetic field change at the bimodal peaks and/or a difference between the magnitudes of the magnetic field change at the bimodal peaks. In some aspects, the implant finder **101** may calculate the depth and/or orientation of the implant **100** in response to a user input (e.g., button press) received via the user input **135** of the user interface **127**. In some aspects, the implant finder **101** may cause the user interface **127** (e.g., the display **129** of the user interface **127**) to provide the calculated depth and/or orientation of the implant **100** to a user (e.g., a clinician).

(88) In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may calculate the depth and/or orientation (e.g., downward angulation) of the implant **100** using a data library to map the sensor signal generated by the sensor **105** during one or more movements/sweeps of the implant finder **101** to a depth and/or orientation. In some aspects, the library may include sensor signal curves characteristic of particular orientations of the implant **100**. In some aspects, the characteristic curves may have been identified through experimentation. In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may compare the sensor signal generated by the sensor **105** during one or more movements/sweeps of

the implant finder **101** to the characteristic curves. In some aspects, the implant finder **101** may determine that the implant **100** has approximately the implant orientation (e.g., in-plane and/or out-of-plane angulation) associated with the characteristic curve to which the generated sensor signal is most similar. In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may calculate the depth (or an estimated depth range) using the magnitude of the sensor signal generated during the movement/sweep (e.g., at one or more peaks thereof).

(89) In some aspects, if the implant finder **101** determines that the sensor signal generated by the sensor **105** during movement of the implant finder **101** is indicative of in-plane angulation (e.g., indicative that the movement/sweep direction is diagonal with respect to the longitudinal axis of the implant **100**), the third step of the second process may include the implant finder **101** (e.g., the computer **106** of the implant finder **101**) determining suggested modification to the movement/sweep direction. In some aspects, the suggested modification may be a particular angle with respect to set direction (e.g., the long axis of the arm) or an angle change (e.g., $+5^\circ$ or -15° with respect to the current/most-recent movement direction. In some aspects, the implant finder **101** may cause the user interface **127** (e.g., the display **129** of the user interface **127**) to provide the suggested modification to the movement/sweep direction to a user. In some aspects, the movement/sweep direction may be adjusted until the generated sensor signal looks as close as the baseline as possible. In some aspects, if one peak is still present in the generated sensor signal, the one peak is most likely due to out-of-plane angulation, which cannot be corrected by adjusting sweep direction.

(90) In some aspects, as shown in FIG. **15C**, the third step of the second process may include marking an implant edge (e.g., using fittings on the outer edge of the magnetic field generator **103**) and/or marking an incision location (e.g., using the incision marking tool **113**).

(91) In some aspects, a third process may be used to locate the implant **100**. In some aspects, the third process may include a first step of finding an approximate location of the longitudinal axis of the implant **100**. In some aspects, as shown in FIG. **16A**, the first step may include moving the implant finder **101** along the skin surface **154** and searching for any signal. In some aspects, the implant **100** may be located in the region where the signal is found.

(92) In some aspects, as shown in FIG. **16B**, the third process may include a second step in which a template **1601** (e.g., a template sticker) is placed over the region where the signal was found. In some aspects, as shown in FIG. **16B**, the template **1601** may include markings (e.g., numbers) that act as way-points identifying a course of movement/sweeping for the implant finder **101**. For example, in some aspects, the template **1601** may identify a course, for example and without limitation, from waypoints **9** to **3**, **4** to **10**, **11** to **5**, **6** to **12**, **1** to **7**, and **8** to **2**. In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may cause the user interface **127** (e.g., the display **129**) to provide instructions for the course of movement/sweeping of the implant finder **101** using the template **1601**. In some aspects, the second step of the third process may include moving the implant finder **101** along the course identified by the template **1601**. In some aspects, the second step of the third process may end with movement of the implant finder **101** to a final position on top of the template **1601**.

(93) In some aspects, the third process may include a third step in which the implant finder **101** (e.g., the computer **106** of the implant finder **101**) calculates the depth and/or orientation (e.g., in-plane angulation and/or out-of-plane angulation) of the implant **100**. In some aspects, the implant finder **101** may calculate the depth and/or orientation of the implant **100** using the sensor signal generated by the sensor **105**, which is indicative of magnetic field changes caused by the magnetic material **124** of the implant **100**, as the implant finder **101** is moved along the course identified by the template **1601**. In some aspects, the implant finder **101** (e.g., the computer **106** of the implant finder **101**) may calculate the depth and/or orientation of the implant **100** using the data library to map the sensor signal generated by the sensor **105** as the implant finder **101** is moved along the course identified by the template **1601** to a depth and/or orientation. In some aspects, the implant

finder **101** may calculate the depth and/or orientation of the implant **100** in response to a user input (e.g., button press) received via the user input **135** of the user interface **127**. In some aspects, the implant finder **101** may cause the user interface **127** (e.g., the display **129** of the user interface **127**) to provide the calculated depth and/or orientation of the implant **100** to a user (e.g., a clinician). (94) In some aspects, as shown in FIG. **16C**, the implant finder **101** may display an implant image **704** and/or a template image **1603** on the screen **702** of the display **129**. In some aspects, as shown in FIG. **16C**, the location of the implant image **704** relative to the template image **1603** on the screen **704** of the display **129** may correspond to a detected location of the implant **100** relative to the template **1601**. In some aspects, as shown in FIG. **16C**, the implant image **704** may have an orientation that corresponds to the calculated orientation of the implant **100**. In some aspects, the implant image **704** may show in-plane angulation and/or out-of-plane (e.g., downward) angulation of the calculated orientation. In some aspects, the implant image **704** may additionally or alternatively show the calculated depth.

(95) In some aspects, the third step of the third process may additionally or alternatively include the implant finder **101** (e.g., the computer **106** of the implant finder **101**) calculating one or more edges of the implant **100**, a center of the implant **100**, and/or an incision location. In some aspects, the implant finder **101** may calculate the one or more implant edges, the implant center, and/or the incision location using the sensor signal generated by the sensor **105**, which is indicative of magnetic field changes caused by the magnetic material **124** of the implant **100**, as the implant finder **101** is moved along the course identified by the template **1601**. In some aspects, the implant finder **101** may calculate the one or more implant edges and/or the incision location using the calculated depth and/or orientation of the implant **100**. In some aspects, the implant finder **101** may cause the user interface **127** (e.g., the display **129** of the user interface **127**) to provide the calculated one or more implant edges, implant center, and/or incision location to a user (e.g., a clinician). In some aspects, as shown in FIG. **16C**, the implant finder **101** may display the calculated one or more implant edges, implant center, and/or incision location on the screen **702** of the display **129** as coordinates.

(96) FIG. **17** is a flow chart illustrating a process **1700** for locating an implant **100** including magnetic material **124** according to some aspects. In some aspects, the process **1700** may include a step **1702** of using a magnetic field generator **103** to generate a magnetic field. In some aspects, the magnetic field generator **103** may include one or more magnets **109**. In some aspects, the magnetic field generator **103** may include a cylindrical magnet **109** (e.g., a hollow cylindrical magnet). In some alternative aspects, the magnetic field generator **103** may include two or more magnets **109** (e.g., four or six magnets). In some aspects, the magnetic field generator **103** may include a housing **107** configured to hold the two or more magnets **109**. In some aspects, the magnetic field generated by the magnetic field generator **103** may be a substantially uniform magnetic field.

(97) In some aspects, the process **1700** may include a step **1704** of using a sensor **105** to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field. In some aspects, the magnetic material **124** of the implant **100** may cause changes to the magnetic field as the sensor **105** is moved over the implant **100**. In some aspects, the magnetic field may be substantially symmetric about a longitudinal axis at the center of the magnetic field generator **103**. In some aspects, the sensor **105** may be located along or offset from the longitudinal axis at the center of the magnetic field generator **103**.

(98) In some aspects, the process **1700** may include a step **1706** of using a computer **106** to detect a location of the implant **100** based on the sensor signal. In some aspects, the step **1706** may include the computer **106** causing the user interface **127** to provide an output indicative of the sensor signal. In some aspects, the user (e.g., a clinician) may use the output sensor signal to detect the location of the implant **100**. In some aspects, the step **1706** may include the computer **106** causing the user interface **127** to provide indications of when a derivative of the changes in the magnetic field equals zero during movement of the sensor **105** across a longitudinal axis of the implant **100**.

In some aspects, the indications of the derivative of the changes in the magnetic field equaling zero during movement of the sensor **105** across the longitudinal axis of the implant **100** may be indicative of the location of the implant **100**.

(99) In some aspects, the step **1706** may additionally or alternatively include the computer **106** determining edges of the magnetic material **124** of the implant **100** based on locations of bimodal peaks in the changes in the magnetic field during movement of the sensor **105** along a longitudinal axis of the implant **100**. In some aspects, the computer **106** may determine edges of the implant **100** based on the determined edges of the magnetic material **124** of the implant **100** and one of more offsets between edges of the implant **100** and edges of the magnetic material **124** of the implant **100**. In some aspects, the computer **106** may be configured to calculate a derivative of the changes in the magnetic field and to use the calculated derivative to detect the location of the implant **100**. In some aspects, the computer **106** may be configured to determine edges of the magnetic material **124** of the implant **100** based on locations where the derivative of the changes in the magnetic field equals zero during movement of the sensor **105** along a longitudinal axis of the implant **100**.

(100) In some aspects, the step **1706** may additionally or alternatively include the computer **106** determining a midline of the implant **100**. In some aspects, the computer **106** may be configured to determine the midline based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor **105** across a longitudinal axis of the implant **100**.

(101) In some aspects, the process **1700** may include a step in which the computer **106** determines a depth of the implant **100** based on a magnitude of the change in the magnetic field at the bimodal peaks in the changes in the magnetic field during the movement of the sensor **105** along the longitudinal axis of the implant **100**. In some aspects, the computer **106** may determine a depth of the implant **100** based on magnitudes of the changes in the magnetic field at the locations where the derivative of the changes in the magnetic field equals zero during the movement of the sensor **105** along the longitudinal axis of the implant **100**.

(102) In some aspects, the process **1700** may additionally or alternatively include a step in which the computer **106** determines an orientation of the implant **100**. In some aspects, the computer **106** may determine the orientation based on a difference between magnitudes of the change in the magnetic field at the bimodal peaks in the changes in the magnetic field during the movement of the sensor **105** along the longitudinal axis of the implant **100**. In some aspects, the computer **106** may determine an orientation of the implant **100** based on a difference between magnitudes of the changes in the magnetic field at the locations where the derivative of the changes in the magnetic field equals zero during the movement of the sensor **105** along the longitudinal axis of the implant **100**.

(103) In some aspects, the process **1700** may additionally or alternatively include computer **106** causing a display **129** to display an indication of the detected location of the implant **100**. In some aspects, the indication of the detected location of the implant **100** may include an implant image **704**, and a location of the implant image **704** on a screen **702** of the display **129** relative to a point **706** on the screen **702** of the display **129** may correspond to the detected location of the implant **100** relative to the sensor **105**. In some aspects, the implant image **704** may have an orientation that corresponds to a detected orientation of the implant **100**.

(104) In some aspects in which the implant finder **101** includes the position detector **137** configured to generate a location signal indicative of a location of the sensor **105**, the process **1700** may additionally or alternatively include a step in which the computer **106** uses the sensor signal and the location signal (e.g., a motion signal generated by a motion detector of the position detector **137** that is indicative of movement of the sensor **105**) to generate a map of sensor signals at different locations of the sensor **105**. In some aspects, generating a map of sensor signals at different locations of the sensor may include, for example and without limitation, measuring the sensor signal at each of two or more different locations of the sensor **105** and storing the measured sensor signals along with identifications of the locations at which the sensor signals were

measured. In some aspects, generating a map of sensor signals at different locations of the sensor **105** may include generating a visualization of the measured sensor signals at the different locations of the sensor **105**.

(105) In some aspects, the process **1700** may include an optional step of removing the implant **100**. In some aspects, removing the implant **100** may include making an incision at an identified edge of the implant **100**. In some aspects, removing the implant **100** may include grabbing (e.g., using forceps) the implant **100** and pulling the implant **100** out of the body through the incision.

(106) FIG. **18** is a flow chart illustrating a process **1800** for locating an implant **100** including magnetic material **124** according to some aspects. In some aspects, the process **1800** may include a step **1802** of moving an apparatus (e.g., implant finder **101**) including a magnetic field generator **103** and a sensor **105** across a longitudinal axis of the implant **100**. In some aspects, the magnetic material **124** of the implant **100** may cause changes in a magnetic field generated by the magnetic field generator **103** as the apparatus is moved across the longitudinal axis of the implant **100**, and the sensor **105** may detect changes in the magnetic field. In some aspects, the process **1800** may include a step **1804** of determining a midline of the implant **100** based on a location where the changes in the magnetic field are greatest as the apparatus is moved across the longitudinal axis of the implant **100**. In some aspects, the process **1800** may include a step **1806** of moving the apparatus along the determined midline of the implant **100**. In some aspects, the process **1800** may include a step **1808** of determining edges of the magnetic material **124** of the implant **100** based on locations of bimodal peaks in the changes in the magnetic field. In some aspects, the process **1800** may include an optional step of using an incision marking tool **113** of the apparatus to mark an incision location for removing the implant **100**. In some aspects, the process **1800** may include an optional step of removing the implant **100**. In some aspects, removing the implant **100** may include making an incision at the incision location. In some aspects, removing the implant **100** may include grabbing (e.g., using forceps) the implant **100** and pulling the implant **100** out of the body through the incision.

(107) In some aspects, as shown in FIG. **19**, the implant **100** may include more than one analyte indicator **104** (e.g., first analyte indicator **104a** and second analyte indicator **104b**). In some aspects, as shown in FIG. **19**, the implant **100** may include a charge storage device **202** (e.g., a battery). In some aspects, the charge storage device **202** may be a battery with a metal housing (e.g., a titanium housing). In some aspects, the charge storage device **202** may be attached to the housing **102**. In some aspects, a coupler **324** may attach the housing **102** and the charge storage device **202**. In some aspects, the coupler **324** may be between the housing **102** and the charge storage device **202**. In some aspects, as shown in FIG. **19**, the coupler **324** may include one or more supports **232** (e.g., reinforcement rods, bars, or beams), which may be attached to and/or integral with the coupler **324**. In some aspects, the implant **100** may include first and second electrically conductive connectors, which connect positive and negative terminals, of the charge storage device **202** to the circuitry of the implant **100**. In some aspects, as shown in FIG. **19**, the circuitry of the implant **100** may extend away from the charge storage device **202** along the longitudinal axis of the charge storage device **202**. In some aspects, the circuitry of the implant **100** may include the inductor **120**, which may include the conductor **122** and the magnetic material **124** in form of a magnetic core. In some aspects, as shown in FIG. **19**, the charge storage device **202** may include a first edge at one end of the implant **100** and a second edge adjacent to the coupler **324** and/or housing **102**. In some aspects, the second edge of the of the charge storage device **202** may be located in a central region of the implant **100**.

(108) In some aspects, the implant **100** may include one or more drug eluting polymer matrices **730** and **732**. In some aspects, the one or more drug eluting polymer matrices **730** and **732** may be in or on portions of the exterior surface of the housing **102** of the implant **100**). In some aspects, one or more therapeutic agents may be dispersed within the one or more drug eluting polymer matrices **730** and **732**. In some aspects, the one or more therapeutic agents may reduce or stop the migration

of neutrophils from entering the space in which the implant **100** has been implanted and, thus, reduce or stop the production of hydrogen peroxide and fibrotic encapsulation. Accordingly, in some aspects, the one or more therapeutic agents may reduce deterioration of the one or more analyte indicators **104** (e.g., first analyte indicator **104a** and second analyte indicator **104b**). In some aspects, the one or more therapeutic agents, which may be dispersed within the one or more drug eluting polymer matrices **730** and **732**, may include one or more anti-inflammatory drugs, such as, for example, non-steroidal anti-inflammatory drug (e.g., acetylsalicylic acid (aspirin) and/or isobutylphenylpropanoic acid (ibuprofen)). In some aspects, the one or more therapeutic agents dispersed within the one or more drug eluting polymer matrices **730** and **732** may include one or more glucocorticoids. In some aspects, the one or more therapeutic agents may include one or more of dexamethasone, triamcinolone, betamethasone, methylprednisolone, beclometasone, fludrocortisone, derivatives thereof, and analogs thereof. In some aspects, the one or more therapeutic agents may reduce the production of hydrogen peroxide by neutrophils and macrophages.

(109) In some aspects (e.g., some aspects in which the implant **100** includes a charge storage device **202**), as shown in FIG. **20**, the magnetic field generator **103** of the implant finder **101** may be an electromagnetic field generator **103'** configured to generate an electromagnetic field. In some aspects, the electromagnetic field may include low frequency waves (e.g., very low frequency (VLF) waves). In some aspects, the low frequency waves may have a frequency, for example and without limitation, within a range between 3 kHz and 30 kHz. In some aspects, the low frequency waves may have a frequency, for example and without limitation, within a range between 5 kHz and 15 kHz. In some aspects, the sensor **105** of the implant finder **101** may be configured to detect changes in the electromagnetic field and to generate a sensor signal indicative of the changes in the electromagnetic field. In some aspects, at least the charge storage device **202** of the implant **100** may cause changes to the electromagnetic field as the sensor **105** is moved over the implant **100**. In some aspects, the computer **106** of the implant finder **101** may be configured to use the sensor signal to detect a location of the implant **100**.

(110) In some electromagnetic field aspects, as shown in FIGS. **21A** and **21B**, the electromagnetic field generator **103'** may include one or more electromagnets **109'**. In some aspects, the one or more electromagnets **109'** may be, for example and without limitation, a coil of wire wound around a magnetic core made from a ferromagnetic or ferrimagnetic material. In some aspects, the electromagnetic field generator **103'** may supply an alternating current to the one or more electromagnets **109'**. The alternating current may have, for example and without limitation, a frequency within a range of between 3 kHz and 30 kHz or, more particularly, within a range between 5 kHz and 15 kHz. In some aspects, as shown in FIG. **21C**, the electromagnetic field generator **103'** may include a housing **107**, and the housing **107** may be configured to hold the one or more electromagnets **109**. However, a housing **107** is not required, and, in some alternative aspects (e.g., some aspects in which the magnetic field generator **103'** consists of only a single electromagnet **109'**), the electromagnetic field generator **103'** may not include a housing **107**. In some aspects, as shown in FIGS. **21A** and **21B**, the sensor **105** may be located on a longitudinal axis at the center of the magnetic field generator **103'**. However, this is not required, and, in some alternative aspects, the sensor **105** is not located on a longitudinal axis at the center of the electromagnetic field generator **103'**. For example, the sensor **105** may be located adjacent to the electromagnet **109'** of the electromagnetic field generator **103'** (as opposed to at the center of one or more electromagnets **109'** of the electromagnetic field generator **103'**). See FIG. **3A**.

(111) FIG. **22** is a flow chart illustrating a process **2200** for locating an implant **100** including a charge storage device **202** according to some aspects. In some aspects, the process **2200** may include a step **2202** of using an electromagnetic field generator **103'** to generate an electromagnetic field. In some aspects, the electromagnetic field generator **103'** may include one or more electromagnets **109'**. In some aspects, the electromagnetic field generator **103'** may include a

cylindrical electromagnet **109'** (e.g., a hollow cylindrical magnet). In some alternative aspects, the electromagnetic field generator **103'** may include two or more electromagnets **109'** (e.g., four or six electromagnets). In some aspects, the electromagnetic field generator **103'** may include a housing **107** configured to hold the one or more electromagnets **109'**.

(112) In some aspects, the process **2200** may include a step **2204** of using a sensor **105** to detect changes in the electromagnetic field and to generate a sensor signal indicative of the changes in the electromagnetic field. In some aspects, at least the charge storage device **202** of the implant **100** may cause changes to the electromagnetic field as the sensor **105** is moved over the implant **100**. In some aspects, the electromagnetic field may be substantially symmetric about a longitudinal axis at the center of the electromagnetic field generator **103'**. In some aspects, the sensor **105** may be located along or offset from the longitudinal axis at the center of the electromagnetic field generator **103'**.

(113) In some aspects, the process **2200** may include a step **2206** of using a computer **106** to detect a location of the implant **100** based on the sensor signal. In some aspects, the step **2206** may include the computer **106** causing the user interface **127** to provide an output indicative of the sensor signal. In some aspects, the user (e.g., a clinician) may use the output sensor signal to detect the location of the implant **100**. In some aspects, the step **2206** may include the computer **106** causing the user interface **127** to provide indications of when a derivative of the changes in the electromagnetic field equals zero during movement of the sensor **105** across a longitudinal axis of the implant **100**. In some aspects, the indications of the derivative of the changes in the electromagnetic field equaling zero during movement of the sensor **105** across the longitudinal axis of the implant **100** may be indicative of the location of the implant **100**.

(114) In some aspects, the step **2206** may additionally or alternatively include the computer **106** determining edges of the charge storage device **202** of the implant **100** based on locations of bimodal peaks in the changes in the electromagnetic field during movement of the sensor **105** along a longitudinal axis of the implant **100**. In some aspects, the computer **106** may determine one or more edges of the implant **100** based on the determined edges of the charge storage device **202** of the implant **100**. In some aspects, the computer **106** may be configured to calculate a derivative of the changes in the electromagnetic field and to use the calculated derivative to detect the location of the implant **100**. In some aspects, the computer **106** may be configured to determine edges of the charge storage device **202** of the implant **100** based on locations where the derivative of the changes in the magnetic field equals zero during movement of the sensor **105** along a longitudinal axis of the implant **100**.

(115) In some aspects, the magnitude of the bimodal peak in the changes in the electromagnetic field at the first edge of the charge storage device **202**, which is at an edge of the implant **100**, may be greater than the magnitude of the bimodal peak in the changes in the electromagnetic field at the second edge of the charge storage device **202**, which is adjacent to the coupler **324** and/or housing **102**. In some aspects, the magnetic material **124** in the housing **102** of the implant **100** may cause the magnitude of the bimodal peak in the changes in the electromagnetic field at the second edge of the charge storage device **202** to be lower than the magnitude of the bimodal peak in the changes in the electromagnetic field at the first edge of the charge storage device **202**. In some aspects, the computer **106** may be configured to determine which of the bimodal peaks in the changes in the electromagnetic field has a greater magnitude. In some aspects, the computer **106** may be configured to determine the first edge of charge storage device **202** to be at the location of the bimodal peak having the greater magnitude. In some aspects, the computer **106** may determine one edge of the implant **100** to be at the first edge of the charge storage device **202**. In some aspects, the incision location for removing the implant **100** may be at or near the location of the bimodal peak in the changes in the electromagnetic field having the greater magnitude.

(116) In some aspects, the computer **106** may be configured to additionally or alternatively determine a second edge of the implant **100**. In some aspects, the computer **106** may determine the

determine the second edge of the implant **100** using the location of the second edge of the charge storage device **202** (e.g., the location of the bimodal peak in the changes in the electromagnetic field having the lower magnitude) and an offset between the second edge of the charge storage device **202** and an end of the housing **102**. In some aspects, the second edge of the charge storage device **202** may be the edge of the charge storage device **202** that is adjacent to the coupler **324** and/or housing **102**.

(117) In some aspects, the step **2206** may additionally or alternatively include the computer **106** determining a midline of the implant **100**. In some aspects, the computer **106** may be configured to determine the midline based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor **105** across a longitudinal axis of the implant **100**.

(118) In some aspects, the process **2200** may include a step in which the computer **106** determines a depth of the implant **100** based on a magnitude of the change in the electromagnetic field at the bimodal peaks in the changes in the electromagnetic field during the movement of the sensor **105** along the longitudinal axis of the implant **100**. In some aspects, the computer **106** may determine a depth of the implant **100** based on magnitudes of the changes in the electromagnetic field at the locations where the derivative of the changes in the electromagnetic field equals zero during the movement of the sensor **105** along the longitudinal axis of the implant **100**.

(119) In some aspects, the process **2200** may additionally or alternatively include a step in which the computer **106** determines an orientation of the implant **100**. In some aspects, the computer **106** may determine the orientation based on a difference between magnitudes of the change in the electromagnetic field at the bimodal peaks in the changes in the electromagnetic field during the movement of the sensor **105** along the longitudinal axis of the implant **100** (e.g., after taking into account the expected difference between the magnitudes at the bimodal peaks due to the second edge of the charge storage device **202** being closer to the magnetic material **124** in the housing **102**). In some aspects, the computer **106** may determine an orientation of the implant **100** based on a difference between magnitudes of the changes in the electromagnetic field at the locations where the derivative of the changes in the electromagnetic field equals zero during the movement of the sensor **105** along the longitudinal axis of the implant **100** (e.g., after taking into account the reduction in the magnitude of the bimodal peak in the changes in the electromagnetic field at the second edge of the charge storage device **202**, which may be caused by the proximity of the second edge of the charge storage device **202** to the magnetic material **124**).

(120) In some aspects, the process **2200** may additionally or alternatively include computer **106** causing a display **129** to display an indication of the detected location of the implant **100**. See, e.g., FIGS. **1** and **7A-7F**. In some aspects, the indication of the detected location of the implant **100** may include an implant image **704**, and a location of the implant image **704** on a screen **702** of the display **129** relative to a point **706** on the screen **702** of the display **129** may correspond to the detected location of the implant **100** relative to the sensor **105**. In some aspects, the implant image **704** may have an orientation that corresponds to a detected orientation of the implant **100**.

(121) In some aspects in which the implant finder **101** includes the position detector **137** configured to generate a location signal indicative of a location of the sensor **105**, the process **2200** may additionally or alternatively include a step in which the computer **106** uses the sensor signal and the location signal (e.g., a motion signal generated by a motion detector of the position detector **137** that is indicative of movement of the sensor **105**) to generate a map of sensor signals at different locations of the sensor **105**. In some aspects, generating a map of sensor signals at different locations of the sensor may include, for example and without limitation, measuring the sensor signal at each of two or more different locations of the sensor **105** and storing the measured sensor signals along with identifications of the locations at which the sensor signals were measured. In some aspects, generating a map of sensor signals at different locations of the sensor **105** may include generating a visualization of the measured sensor signals at the different locations of the sensor **105**.

(122) In some aspects, the process **2200** may include an optional step of removing the implant **100**. In some aspects, removing the implant **100** may include making an incision at an identified edge of the charge storage device **202**. In some aspects, removing the implant **100** may include grabbing (e.g., using forceps) the charge storage device **202** of the implant **100**, which may be stronger than the housing **102** of the implant **100**, and pulling the implant **100** out of the body through the incision.

(123) FIG. **23** is a flow chart illustrating a process **2300** for locating an implant **100** including a charge storage device **202** according to some aspects. In some aspects, the process **2300** may include a step **2302** of moving an apparatus (e.g., implant finder **101**) including an electromagnetic field generator **103'** and a sensor **105** across a longitudinal axis of the implant **100**. In some aspects, at least the charge storage device **202** of the implant **100** may cause changes in an electromagnetic field generated by the electromagnetic field generator **103'** as the apparatus is moved across the longitudinal axis of the implant **100**, and the sensor **105** may detect changes in the electromagnetic field. In some aspects, the process **2300** may include a step **2304** of determining a midline of the implant **100** based on a location where the changes in the electromagnetic field are greatest as the apparatus is moved across the longitudinal axis of the implant **100**. In some aspects, the process **2300** may include a step **2306** of moving the apparatus along the determined midline of the implant **100**. In some aspects, the process **2300** may include a step **2308** of determining edges of the charge storage device **202** of the implant **100** based on locations of bimodal peaks in the changes in the electromagnetic field. In some aspects, the process **2300** may include an optional step of using an incision marking tool **113** of the apparatus to mark an incision location for removing the implant **100**. In some aspects, the incision location may be at or near the location of the bimodal peak in the changes in the electromagnetic field that has the greater magnitude. In some aspects, the process **2300** may include an optional step of removing the implant **100**. In some aspects, removing the implant **100** may include making an incision at the incision location. In some aspects, removing the implant **100** may include grabbing (e.g., using forceps) the charge storage device **202** of the implant **100**, which may be stronger than the housing **102** of the implant **100**, and pulling the implant **100** out of the body through the incision.

(124) While the subject matter of this disclosure has been described and shown in considerable detail with reference to certain illustrative embodiments, including various combinations and sub-combinations of features, those skilled in the art will readily appreciate other embodiments and variations and modifications thereof as encompassed within the scope of the present disclosure. Moreover, the descriptions of such embodiments, combinations, and sub-combinations is not intended to convey that the claimed subject matter requires features or combinations of features other than those expressly recited in the claims. Accordingly, the scope of this disclosure is intended to include all modifications and variations encompassed within the spirit and scope of the following appended claims.

Claims

1. An apparatus for locating an implant comprising magnetic material in a living animal, the apparatus comprising: a magnetic field generator configured to generate a magnetic field; a sensor configured to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field, wherein the magnetic material of the implant causes changes to the magnetic field as the sensor is moved over the implant; and a computer configured to use the sensor signal to detect a location of the implant, wherein, in detecting the location of the implant, the computer is configured to determine edges of the magnetic material of the implant based on locations of bimodal peaks in the changes in the magnetic field during movement of the sensor along a longitudinal axis of the implant.
2. The apparatus of claim 1, wherein the magnetic field generator comprises a cylindrical magnet.

3. The apparatus of claim 2, wherein the cylindrical magnet is hollow.
4. The apparatus of claim 1, wherein the magnetic field generator comprises two or more magnets.
5. The apparatus of claim 4, wherein the magnetic field generator further comprises a housing configured to hold the two or more magnets.
6. The apparatus of claim 4, wherein the magnetic field generator comprises four magnets.
7. The apparatus of claim 4, wherein the magnetic field generator comprises six magnets.
8. The apparatus of claim 1, wherein the magnetic field generator comprises one or more permanent magnets.
9. The apparatus of claim 1, wherein the magnetic field generator comprises one or more electromagnets.
10. The apparatus of claim 1, wherein the magnetic field generated by the magnetic field generator is a substantially uniform magnetic field.
11. The apparatus of claim 1, wherein the magnetic field is substantially symmetric about a longitudinal axis at a center of the magnetic field generator.
12. The apparatus of claim 11, wherein the sensor is located along or offset from the longitudinal axis at the center of the magnetic field generator.
13. The apparatus of claim 1, wherein the magnetic field is non-uniform and/or asymmetric about a longitudinal axis at a center of the magnetic field generator.
14. The apparatus of claim 1, wherein the computer is configured to determine edges of the implant based on the determined edges of the magnetic material of the implant and one of more off sets between edges of the implant and edges of the magnetic material of the implant.
15. The apparatus of claim 1, wherein the computer is configured to determine a depth of the implant based on a magnitude of the change in the magnetic field at the bimodal peaks in the changes in the magnetic field during the movement of the sensor along the longitudinal axis of the implant.
16. The apparatus of claim 1, wherein the computer is configured to determine an orientation of the implant based on a difference between magnitudes of the change in the magnetic field at the bimodal peaks in the changes in the magnetic field during the movement of the sensor along the longitudinal axis of the implant.
17. The apparatus of claim 1, wherein the computer is further configured to use one or more sensor signals to detect an orientation of the implant.
18. The apparatus of claim 1, further comprising a display, and the computer is configured to cause the display to display an indication of the detected location of the implant.
19. The apparatus of claim 18, wherein the display is located above the sensor.
20. The apparatus of claim 18, wherein the indication of the detected location of the implant comprises an implant image, and a location of the implant image on a screen of the display relative to a point on the screen of the display corresponds to the detected location of the implant relative to the sensor.
21. The apparatus of claim 20, wherein the implant image has an orientation that corresponds to a detected orientation of the implant.
22. The apparatus of claim 1, further comprising an incision marking tool configured to identify an incision location for removing the implant.
23. The apparatus of claim 1, further comprising a position detector configured to generate a location signal indicative of a location of the sensor on a skin surface.
24. The apparatus of claim 23, wherein the position detector comprises a motion detector configured to detect movement of the sensor and to generate a motion signal indicative of the detected movement of the sensor, wherein the location signal comprises the motion signal.
25. The apparatus of claim 23, wherein the computer is configured to use the sensor signal and the location signal to generate a map of sensor signals at different locations of the sensor on the skin surface.

26. An apparatus for locating an implant comprising magnetic material in a living animal, the apparatus comprising: a magnetic field generator configured to generate a magnetic field; a sensor configured to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field, wherein the magnetic material of the implant causes changes to the magnetic field as the sensor is moved over the implant; and a computer configured to use the sensor signal to calculate a derivative of the changes in the magnetic field and to use the calculated derivative to detect a location of the implant, wherein the computer is configured to determine edges of the magnetic material of the implant based on locations where the derivative of the changes in the magnetic field equals zero during movement of the sensor along a longitudinal axis of the implant.
27. The apparatus of claim 26, wherein the computer is configured to determine a depth of the implant based on magnitudes of the changes in the magnetic field at the locations where the derivative of the changes in the magnetic field equals zero during the movement of the sensor along the longitudinal axis of the implant.
28. The apparatus of claim 26, wherein the computer is configured to determine an orientation of the implant based on a difference between magnitudes of the changes in the magnetic field at the locations where the derivative of the changes in the magnetic field equals zero during the movement of the sensor along the longitudinal axis of the implant.
29. An apparatus for locating an implant comprising magnetic material in a living animal, the apparatus comprising: a magnetic field generator configured to generate a magnetic field; a sensor configured to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field, wherein the magnetic material of the implant causes changes to the magnetic field as the sensor is moved over the implant; and a computer configured to use the sensor signal to calculate a derivative of the changes in the magnetic field and to use the calculated derivative to detect a location of the implant, wherein the computer is configured to determine a midline of the implant based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor across a longitudinal axis of the implant.
30. A method for locating an implant comprising magnetic material in a living animal, the method comprising: using a magnetic field generator to generate a magnetic field; using a sensor to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field, wherein the magnetic material of the implant causes changes to the magnetic field as the sensor is moved over the implant; and using a computer to detect a location of the implant based on the sensor signal, wherein detecting the location of the implant comprises determining edges of the magnetic material of the implant based on locations of bimodal peaks in the changes in the magnetic field during movement of the sensor along a longitudinal axis of the implant.
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